

A terminal on Γ in Jochum's formulation — derivation report

Status: §6 has been tested. It gets the physics right and the gauge wrong. §§1–5 are worked and stand. §6's construction reproduces the correct terminal impedance at every frequency — and is **as tree-dependent as the tree-cotree solve it was meant to replace**. §10 has the measurement and the diagnosis. Nothing is implemented in C++.

Why this document exists: `tools/gauge_spike/RESULTS_SPIKEC.md` milestone 2 found that Jochum, Farle & Dyczij-Edlinger's formulation — the only one in the literature carrying a materially-weighted gauge interface condition at a conductor/insulator boundary, which is exactly where our failing port sits — has **nowhere to put a port**. This works out why, and what would have to change.

1. The formulation, in our convention

Jochum writes $e^{+ik_0 \cdot}$ with a scaled potential; translated to ours ($e^{+j\omega t}$), $E = -j\omega A - \nabla\Phi$, with $\kappa = \sigma + j\omega\epsilon$ and $v = 1/\mu$:

$$\begin{aligned} A &= A_c + \nabla\psi, & B &= \nabla \times A_c, & (15), (16) \\ E &= -\nabla V - j\omega(A_c + \nabla\psi) & & & (17) \end{aligned}$$

Three unknowns: A_c on cotree edges, ψ nodal, V nodal. A_c lives in the reduced space of the **inexact Helmholtz splitting** (14), realized discretely by tree-cotree — so the tree is still there, but it decomposes the space rather than fixing the gauge.

In the lossy sub-domain Ω_C :

$$\begin{aligned} \nabla \times (v \nabla \times A_c) + \kappa [j\omega(A_c + \nabla\psi) + \nabla V] &= J_i & \text{Ampère} & (18a) \\ \nabla \cdot [\kappa(A_c + \nabla\psi)] &= 0 & \text{GAUGE} & (18b) \end{aligned}$$

In the lossless sub-domain Ω_N :

$$\begin{aligned} \nabla \times (v \nabla \times A_c) + j\omega\epsilon [j\omega(A_c + \nabla\psi) + \nabla V] &= J_i & \text{Ampère} & (19a) \\ \nabla \cdot \{\epsilon [j\omega(A_c + \nabla\psi) + \nabla V]\} &= \rho & \text{Gauss} & (19b) \end{aligned}$$

Since $\kappa|_{\Omega_N} = j\omega\epsilon$, (19a) is (18a) with the local κ — one expression covers both. The asymmetry is entirely in the **second** equation: a gauge in the conductor, physics in the insulator.

2. Which equation lives in which row — the fact everything turns on

Jochum tests each equation against a different space (his §5.2):

WEAK FORM	EQUATION TESTED	TEST FUNCTION	PRODUCES THE ROW FOR
(21)	Ampère (18a)	$w_c \in \tilde{Q}H(\text{curl})$	** A_c **
(22)	Ampère (18a)	** $\nabla\psi$ **, $\psi^- \in H^1_o$	** ψ **
(23)	gauge (18b)	$\bar{V} \in H^1_o$	** V **

and states that (22) is a weak form of the **continuity equation**.

So, and this is the crux:

** ψ 's row is continuity. V 's row is the gauge.**

That cross-placement is what makes the off-diagonal blocks transposes and the matrix complex-symmetric — it is a feature, not an accident. But it **inverts the usual A - Φ correspondence**, where a current-driven port's row is the continuity equation and its unknown is the potential.

3. Deriving (32) without a terminal

Test (18b) by $\bar{V}$ over Ω_C and integrate by parts:

$$-\int_{\Omega_C} \kappa(A_c + \nabla\psi) \cdot \nabla\bar{V} + \oint_{\partial\Omega_C} \bar{V} \hat{n} \cdot [\kappa(A_c + \nabla\psi)] = 0$$

Test (19b) by $\bar{V}$ over Ω_N , noting $\epsilon[j\omega(A_c + \nabla\psi) + \nabla V] = -\epsilon E$:

$$-\int_{\Omega_N} (-\epsilon E) \cdot \nabla \bar{V} + \oint_{\partial\Omega_N} \bar{V} \hat{n} \cdot (-\epsilon E) = \int \rho \bar{V}$$

Let $\hat{n}$ point from Ω_C into Ω_N , so $\hat{n}_C = \hat{n}$ and $\hat{n}_N = -\hat{n}$ on Γ . The two surface integrals contribute, on Γ ,

$$\oint_{\Gamma} \bar{V} \{ \hat{n} \cdot [\kappa(A_C + \nabla\psi)]_C + \hat{n} \cdot (\epsilon E)_N \} dS \quad (\star)$$

With $\bar{V}$ single-valued and **free** on Γ , $\bar{V}$ is arbitrary there, so $(\star)$ vanishes only if its integrand does pointwise:

$$\hat{n} \cdot [\kappa(A_C + \nabla\psi)]_C + \hat{n} \cdot (\epsilon E)_N = 0 \quad \text{on } \Gamma \quad (32)$$

This is Jochum's (32), and it is not an extra assumption — it is the condition under which the two sub-domain weak forms assemble into one system with no leftover surface term. It is materially weighted on both sides: $\kappa = \sigma + j\omega\epsilon$ from the conductor, ϵ from the insulator. That weighting is why §12.3 ranks it above the ϵ -blind Coulomb-family conditions.

4. What a terminal is, formally

A current-driven terminal on $\Gamma_T \subseteq \Gamma$ is two statements:

1. **$\bar{V}$ is single-valued on Γ_T** — one unknown $\bar{V}_T$, not one per node. In the discrete setting this is the aggregation $A = P \hat{A} P^T$ (Stysch §6.6.2, [GAUGE_CHOICE.md](#) §11.10.5), which is a congruence and so preserves symmetry.
2. **The total current through Γ_T is prescribed**:

$$I = \int_{\Gamma_T} \hat{n} \cdot (\kappa E)_C dS \quad (33)$$

Statement 1 collapses the test space: on Γ_T , $\bar{V}$ is a single constant rather than an arbitrary function.

5. The obstacle, located exactly

With $\bar{V}$ constant on Γ_T , $(\star)$ no longer forces its integrand to vanish there. It forces only the **integral**:

$$\int_{\Gamma_T} \{ \hat{n} \cdot [\kappa(A_c + \nabla\psi)]_C + \hat{n} \cdot (\epsilon E)_N \} dS = 0 \quad (34)$$

That is one scalar equation, which is right — one equation for one unknown V_T . But now look at what it contains against what we need to impose:

NORMAL FLUX APPEARING

(34), the V_T row $\hat{n} \cdot (\kappa \star A \star)_C$ — the **gauge** flux

(33), the current $\hat{n} \cdot (\kappa \star E \star)_C$ — the **current**

These are different physical objects. A is not E . The gauge equation's natural boundary datum is the normal component of κA , which is a statement about the gauge, and no choice of right-hand side turns it into a current.

And the current flux $\hat{n} \cdot (\kappa E)$ does appear in the weak form — but in the $\star \psi$ row, since (22) is continuity and its surface term is $\oint \hat{n} \cdot (\kappa E) \psi$.

So the terminal's two requirements land in two different rows:

the terminal POTENTIAL we want to read	is	V_T	
the row V_T owns	is	the GAUGE	(34)
the row carrying the CURRENT	is	ψ 's	(continuity)

That is the whole obstacle, and it is structural rather than a matter of assembly detail.

5.1 It explains the measurement exactly

Spike C milestone 2 put the current drive on V 's row — the natural thing to do, and wrong here. Measured consequence: a terminal potential of $-1.149425e-05$ V identical across three trees to $2.4e-14$, with **zero reactance, unchanged from 50 Hz to 50 GHz**, and a V-coupling of $G_{\text{eps}N} = 4.2e-02$ against gauge terms $G_{\text{kap}} = 7.5e+07$ — a factor $5.7e-10$.

Every one of those follows from §5: the drive perturbed the gauge condition instead of injecting current, leaving a decoupled DC conduction problem whose solution is linear in z and therefore exact in P1. The spike did not fail; it measured this.

5.2 Why his paper never hit it

His §6.1 cavity sets $\phi = 0$ and $\phi = 1$ on faces, and §6.2 is a voltage-driven RLC field model. Both are **Dirichlet conditions on V **, which are trivial here — prescribe V and the row disappears. **A voltage port in Jochum's formulation is free; only a current-driven floating terminal is a problem.** That is a useful narrowing: the gap is one port type, not the whole concept.

6. Proposed resolution — aggregate ψ as well as V

Not verified. The dimension count balances and the physics is plausible; that is all that can be claimed so far.

If Γ_T needs two conditions — the gauge (34) and the current (33) — then give it **two unknowns**: aggregate ψ on Γ_T as well as V .

$V_T = V$	on Γ_T	one unknown	row: aggregated gauge	(34)
$\psi_T = \psi$	on Γ_T	one unknown	row: aggregated continuity = I	(33)

Count. With N_T nodes on Γ_T , before: N_T V -unknowns and N_T ψ -unknowns against N_T gauge rows and N_T continuity rows. After: 2 unknowns against 2 rows. Balanced.

****Legitimacy of aggregating ψ .** ψ is the gauge function. Forcing it constant on a surface is a *gauge choice*, not a physical constraint — it is precisely what contracting a surface in the tree does (§15.9), and what $\hat{n} \times A = 0$ achieves on the outer boundary. So it costs nothing physical.

Symmetry. Both aggregations are congruences $P(\cdot)P^T$, applied to the V and ψ blocks. A congruence preserves complex symmetry, so the system should remain symmetric — the property §13.2 identifies as Jochum's main advantage over Chew.

****What V_T then means.** V_T is the terminal potential, read as the answer. ψ_T is a gauge constant on the terminal, carrying no physical meaning — the analogue of $\psi = 0$ on the outer boundary.

6.1 What is not yet established

- **Uniqueness.** Does (33)+(34) determine (V_T, ψ_T) uniquely, or is the pair

singular in some limit? The count balances; that is necessary, not sufficient.

- **Whether (32) survives off Γ_T .** The derivation in §3 assumed $\bar{V}$ free on

all of Γ . On $\Gamma \setminus \Gamma_T$ it still is, so (32) should hold there — but the two regions share an edge, and the rim of Γ_T needs checking.

- **The low-frequency limit.** $\kappa \rightarrow \sigma$ in the conductor and $\kappa \rightarrow 0$ in the

insulator as $\omega \rightarrow 0$. §11.10 measured that an insulator-interior ψ row vanishes at DC, which is why Stysch's §5.5 restricts ψ to the interface. Γ_T is on the interface, so ψ_T should survive — but that is an expectation, not a measurement.

- **Whether I belongs on ψ_T 's row with a factor.** (33) is the current;

(22) is continuity tested by $\nabla\psi$. The aggregated continuity row equals the net current through Γ_T only up to a sign and possibly a scaling from the non-dimensionalization. That needs doing carefully, and `RESULTS_COMPARE.md` records what dropping such a factor costs.

7. How to test it, cheaply

`tools/gauge_spike/spikeC.py` already has everything but this: a validated miniature (milestone 1: R exact to 0.06 %, terminal ϕ moving 3.8 % with the tree), a Jochum assembly (milestone 2), and a terminal aggregation.

The change is to `solve_jochum`: add a second aggregated unknown on the terminal face for ψ , move the drive from V 's row to ψ 's, and re-run. The acceptance test is unchanged and already written:

	MUST BE
R	11.494253 $\mu\Omega$ — the analytic DC value
reactance	present and frequency-dependent, not constant across decades
terminal ϕ across three trees	invariant
symmetry residual	$\sim 1e-16$

Milestone 2's failure signature is specific enough to be diagnostic: if the reactance is again zero and frequency-independent, the drive is still in the gauge row and §6 is wrong.

8. If §6 fails

Two fallbacks, both more expensive:

- **Re-derive (32) with the terminal present from the start**, rather than

aggregating afterwards. (32) came from requiring the surface terms to cancel; with a terminal, the correct statement may be a *modified* interface condition on Γ_T rather than the same one aggregated. This is the honest version of "the derivation", and §6 is the shortcut worth trying first.

- **Take the condition from outside.** A boundary-integral exterior supplies the

relation between A and Φ on a coupling surface from the exterior solution, which is what `GAUGE_CHOICE.md` §0 wants Chew for. That removes the need to invent an interface condition at all, at the cost of the whole BI programme.

9. Sources

Jochum, Farle & Dyczij-Edlinger (2016) , <i>A Symmetric and Low-Frequency Stable Potential Formulation...</i> , in <i>Scientific Computing in Electrical Engineering</i> (SCEE 2014), Mathematics in Industry 23 , 63–71, doi:10.1007/978-3-319-30399-4_7 — APhi_Papers/Jochum_SymmetricLowFreqStablePotentialFormulation.pdf	Equations (14)–(19), (21)–(23), (32); §5.2's test-function assignment, which §2 above turns on; §6's examples, which are voltage-driven and so never meet this problem.
Stysch (2022) , PhD thesis, TU Darmstadt	§6.6.2's aggregation $\mathbf{A} = \mathbf{P} \hat{\mathbf{A}} \mathbf{P}^\top$, which §4 and §6 use for the terminal; §5.5's restriction of ψ to the interface, which §6.1 relies on.
tools/gauge_spike/RESULTS_SPIKEC.md	Milestone 2's measurement, which §5.1 explains.
GAUGE_CHOICE.md §13	The reading of Jochum this builds on.

10. Result: §6 tested

[tools/gauge_spike/spikeC.py](#), milestone 5. Two findings, and the second undoes the first.

10.1 The readout in §6 was wrong, and fixing it gives correct physics

§6 said V_T is the terminal potential. **It is not.** From (17),

$$\mathbf{E} = -\nabla V - j k_o (\mathbf{A}_c + \nabla \psi) = -\nabla (V + j k_o \psi) - j k_o \mathbf{A}_c$$

so the scalar content is ****split between V and ψ ****, and the effective potential — the one that plays the usual ϕ against $\mathbf{A}_c$ as the vector potential — is

$$\phi_{\text{eff}} = V + j k_o \psi \tag{35}$$

Read that way, the construction is **exactly right**:

FREQ	JOCHUM ϕ_{EFF}	TREE-COTREE $A-\Phi$, SAME QUANTITY
50 Hz	1.150074e-05 + 1.043363e-06j	1.150074461e-05 + 1.043363113e-06j
50 kHz	3.011561e-04 + 3.567000e-04j	3.011561e-04 + 3.567000e-04j
50 MHz	1.212789e-03 + 3.528559e-06j	1.212789e-03 + 3.528559e-06j

Identical to every digit printed, at every frequency — **R** right, **reactance present and frequency-dependent**, symmetry residual 3.6e-24. Milestone 2's failure signature is gone: the drive now injects current instead of perturbing the gauge, exactly as §5 predicted it would once moved to ψ 's row.

So §6's *structure* — two aggregated unknowns, current on the continuity row — is correct. Reading V_T alone was the error, and it explains why milestone 5 first reproduced milestone 2's symptom.

10.2 And it is exactly as tree-dependent as what it replaces

	ϕ_{EFF} , BASE→PERMA	BASE→PERMB
50 Hz	3.843e-02	3.533e-02
50 kHz	5.318e-01	5.771e-01
50 MHz	2.823e+02	3.246e+02

The tree–cotree $A-\Phi$ solve moves the same quantity by **3.84e-02 and 3.53e-02**. Identical. And it degrades with frequency, to a factor of 282 at 50 MHz.

This fails the acceptance test. The whole point was a formulation whose terminal potential does not depend on the tree.

10.3 Why — and one hypothesis already falsified

First suspicion: Stysch's §5.5 restriction of ψ to the interface and conductor, which I had carried over. $A = A_c + \nabla\psi$ spans the full edge space only if ψ spans all nodes — cotree is $\#edges - \#nodes + 1$, gradients are $\#nodes - 1$ — so restricting ψ punches a hole in the span, and the missing gradients would be tree-dependent.

Falsified. With ψ on every interior node the numbers are identical to every digit: 3.843e-02 and 3.533e-02. Only the conditioning changed, $1.8e+10 \rightarrow 5.6e+26$, which reproduces `RESULTS_COMPARE.md`'s finding that an insulator-interior ψ wrecks the conditioning.

The remaining explanation, and it is a caution about the whole approach:

**** A_c lives in the cotree space, and that space is defined by the tree.**** Jochum's (14) is an *"inexact Helmholtz splitting"* which he realizes discretely **by tree-cotree**. So the tree is baked into the trial space before the gauge condition ever acts. The gauge then determines ψ given a tree-dependent A_c , and the dependence survives into ϕ_{eff} .

That would make the gauge condition a statement about how A is *split* between A_c and $\nabla\psi$, not about which A the discrete problem selects — which is precisely the distinction Rapetti draws when he says tree gauges *"are not a discretization of the Coulomb gauge and enforce no orthogonality."*

Stated as a limit of this work, not of his paper. My construction may be unfaithful: his examples are voltage-driven with no interior terminal (§5.2), so this configuration is outside what he demonstrates, and the aggregation is mine. What is measured is that **the construction in §6 does not deliver a tree-independent terminal**, and that the cause is not the ψ restriction.

10.4 What this leaves

§6's structure	right — correct R , correct reactance, symmetric, drive in the right row
§6 as a fix for the gauge	wrong — tree-dependence identical to tree–cotree's, worsening with frequency
the ψ restriction as the cause	falsified
likely cause	A_c 's space is tree-defined, so the tree precedes the gauge

If that diagnosis is right, no arrangement of terminals rescues it: the fix would have to stop realizing the Helmholtz splitting with a tree. That points at §8's second fallback — taking the condition from an exterior boundary-integral solution — rather than at any further rearrangement of Jochum's rows.

10.5 The control that would settle it was attempted, and failed

The diagnosis in §10.3 needs one check: is **Jochum without a terminal** tree-independent on this mesh? If his published configuration is also tree-dependent here, my construction is faithful and the conclusion holds. If it is not, the terminal is what broke it, and §6 needs revisiting rather than abandoning.

I built it — his configuration, Dirichlet **V** on faces, no terminal, measuring the dissipated power, which is gauge-invariant — and **it does not work**. Two attempts:

ATTEMPT	SYMPTOM	CAUSE	OUTCOME	
1, ψ on all interior nodes	<code>`max\</code>	<code>V\</code>	<code>= 431 V for a 1 V drive,</code> <code>cond = 7.7e+26`</code>	the system is numerically singular at double precision — the same ψ -everywhere conditioning collapse <code>RESULTS_COMPARE.md</code> measured at <code>5.6e+26</code>
2, ψ restricted per Stysch §5.5	<code>`max\</code>	<code>V\</code>	<code>= 1.000 ✓,</code> <code>cond = 1.8e+10</code> <code>✓, but P =</code> <code>2.4e-19 W</code> <code>against an exact</code> <code>4.35e+04 W`</code>	<code>E ≈ 0</code> inside the conductor, where 1 V across 6 mm should give ~167 V/m still wrong, different bug

Attempt 2 establishes that **the Dirichlet lift was correct all along** — **V** comes out in the right range once the conditioning is fixed. Something else in the voltage-driven assembly or in the **E** reconstruction is wrong, and I have not found it.

The 98–99 % tree-dependence both attempts report is therefore **meaningless**: it is 98 % of a quantity that is numerically zero.

Worth stating plainly: the **current-driven** assembly in §10.1 is right — it reproduces the $A-\Phi$ solve digit-for-digit at three frequencies. The bug is specific to the voltage-driven variant built for this control, which shares the operator but not the right-hand side or the drive.

So §10.3 remains a hypothesis, and the two possibilities it was meant to separate are both still open:

- my construction is faithful, and Jochum's formulation inherits tree-dependence through **A_c** 's tree-defined space; **or**
- my construction is unfaithful in some way the terminal exposes, and his published configuration is fine.

What is **not** in doubt, because it was measured three ways and cross-checked digit-for-digit against the $A-\Phi$ solve: §6 gives the right impedance at every frequency, and it does not remove the tree-dependence. Whichever explanation is right, **§6 is not the fix**.

10.6 Tested directly instead — **A itself moves, not just its split**

The broken control was a proxy. The question it stood for can be asked of the **current-driven** assembly, which demonstrably works (§10.1): solve on two trees and compare **A_c** , **$\nabla\psi$** and their sum.

	BASE → PERMA, 50 HZ	50 KHZ
A_c	8.55e-01	7.85e-01
$\nabla\psi$	8.37e-01	7.86e-01
** A = A_c + $\nabla\psi$ **	8.67e-01	7.85e-01
** B = $\nabla\times A_c$ **	9.71e-01	8.52e-01

A itself moves by 87 %, not merely its split between **A_c** and $\nabla\psi$.** And **B** — physical and gauge-invariant — moves by 97 %, so the two runs are **different physical solutions**, not two gauges of one.

That is the second branch of §10.3: **the cotree space is the culprit, and it is structural**. It also closes the chain. **Φ_eff** reproduced the A-Φ solve digit-for-digit (§10.1); **A** and **B** now reproduce its *pathology* — the same 91 % **B** movement **GAUGE_CHOICE.md** §15.3 measures at a mixed port. **This assembly is the tree–cotree A-Φ system re-parameterized**, and the gauge condition is contributing nothing.

10.6a The assembly was unfaithful — and fixing it changed nothing

Checking §§1–5 against the paper term by term found a real defect. Jochum's (26), the ψ row in **Ω_N**, satisfies

row2|Ω_N = jk_o × row3|Ω_N exactly, since κ|Ω_N = jk_oε

He says so — (28) is *"a wavenumber-scaled version of (19b),* *in accordance with the lack of a gauge in **Ω_N** *** — and instructs that **(26) be replaced by a suitable gauge condition, on the FE level**. My assembly never did: it carried the redundant row, so the matrix was rank-deficient by one per insulator-interior node.

Replacing row 2 in **Ω_N** with $\nabla \cdot (\epsilon A) = 0$ fixes that. **The tree-dependence is unchanged:** 3.843e-02, 5.318e-01, 2.822e+02 — identical to three figures.

So the redundancy was real and was not the cause.

10.6b It is the ψ aggregation, and the count proves it

Across the milestones the tree-dependence tracks one thing only:

	ψ AGGREGATED ON Γ_T ?	REACTANCE	TREE-DEPENDENCE
milestone 2, drive on V 's row	no	absent	2.4e-14
milestone 5, drive on ψ 's row	yes	correct	3.8e-02
milestone 5 + the Ω_N gauge fix	yes	correct	3.8e-02

A = A_c + ∇ψ reaches the whole edge space only if ψ carries one DOF per node. Counted on this mesh:

interior edges	1154	what A must span
cotree (A_c)	1054	
interior nodes	100	
cotree + all interior nodes	1154	COMPLETE
psi AGGREGATED on the terminal	89	12 terminal nodes -> 1
cotree + aggregated psi	1143	SHORT BY 11

****Aggregating ψ removes exactly $N_T - 1 = 11$ gradient directions**, and *which 11 are unreachable is decided by the cotree. That is the tree, back in through the port model.***

So §6's proposal is structurally flawed, for a reason that can be counted rather than argued: **the terminal must not consume gradient degrees of freedom.**

10.6c What that indicates

Impose the total current **without** aggregating ψ — a Lagrange multiplier on Γ_T adds one unknown and one equation while removing none, so the span stays complete. V may still be aggregated, since V is not part of A and its aggregation cannot affect the span; milestone 2 confirms a V -only aggregation is tree-invariant to $2.4e-14$.

That is the next thing to try, and it is a small change to the same bed.

10.7 Status, stated plainly

§6's structure	right — correct impedance at every frequency, symmetric, drive in the right row
§6 as a gauge fix	wrong , and now explained: A is tree-dependent, so the gauge never had a chance
the ψ restriction	falsified as the cause
the cotree space	confirmed as where the tree enters
is this faithful to Jochum?	still unknown , and that is the honest limit of this work

The last row matters. Three attempts have produced **no gauge benefit at all**, which is either a true property of realizing the Helmholtz splitting with a tree — his (14) — or a defect in my construction that his voltage-driven examples would never expose. I cannot separate those, and the control that would have has failed twice.

What is safe to conclude: **nothing in this line has yet produced a terminal potential that survives a change of tree** — but §10.6b now locates the obstacle precisely, and it is **the port model, not the formulation**. The terminal consumes gradient degrees of freedom that

A needs in order to span the edge space, and the cotree fills the gap.

That points **back** at Jochum's rows rather than away from them, and at a terminal that **adds a constraint instead of removing unknowns** (§10.6c). The earlier reading — that this indicted the formulation and favoured a boundary-integral exterior — is **withdrawn**: it rested on §10.6's diagnosis, which §10.6b supersedes.